WHITE PAPER

The Last Mile of ISO 20022

Why Address Data Quality Will Determine Who Wins the 2026 Transition— and the Technology That Makes It Possible

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A position paper on the transition from unstructured to fully structured postal addresses in global financial services: the regulatory mandate, why conventional approaches fail, and the purpose-built technology that provides a practical path forward.

Executive Summary

By November 2026, unstructured postal addresses will be eliminated from SEPA and SWIFT CBPR+ payment messages. This deadline is now less than ten months away. The regulatory trajectory is unambiguous: the European Payments Council, SWIFT, and the BIS Committee on Payments and Market Infrastructures have converged on a clear mandate. Structured addresses are the future. Unstructured formats are being retired.

Yet across the industry, a familiar and concerning pattern is emerging. Most institutions have adopted the ISO 20022 message format. Far fewer have addressed the quality of the address data flowing through those formats. Format migration has been treated as synonymous with data readiness. It is not.

This paper presents six core findings:

1. **The compliance deadline is firm and coordinated.** By 22 November 2026, unstructured addresses will no longer be accepted in EPC SEPA schemes or SWIFT CBPR+ cross-border messages. Only structured or hybrid formats will be permitted, with Town/City Name and Country Code as mandatory minimums.
2. **Format migration is not data readiness.** Industry evidence shows that unstructured or poor-quality address content persists inside new ISO 20022 formats. Switching message schemas without cleansing the underlying data delivers compliance in appearance but not in substance.
3. **The cost of inaction is measurable.** Industry analysis estimates that up to 60% of cross-border payments still require manual intervention due to address-related issues, at an annual cost to the industry of $15–25 billion.
4. **Conventional remediation approaches are inadequate.** Regex-based parsing, generic large language models, postal validation services, and manual consultancy projects each fail to meet the specific requirements of payment address structuring at enterprise scale.
5. **Production-grade automated address resolution technology now exists.** Purpose-built platforms combining deterministic processing, context-aware natural language processing, and deep payments domain knowledge can transform unstructured addresses into fully compliant structured output at sub-50-millisecond latency, with complete audit trails. These platforms are already processing billions of transactions annually across major global institutions.
6. **This is a strategic opportunity, not merely a compliance obligation.** Institutions that invest in genuine data quality—structured at source, validated early, governed consistently—will emerge with durable advantages in operational efficiency, risk management, and analytical capability.

The central argument of this paper is that the structured address transition represents the most significant data quality challenge facing financial services operations today—and the most underestimated strategic opportunity. The technology to solve it exists and is proven.

What follows is the evidence, the analysis of why conventional approaches fall short, and a practical framework for institutions ready to act.

Introduction: A Pattern I Have Seen Before

Over three decades in financial services technology, I have observed a pattern that repeats with remarkable consistency. A major industry-wide technology migration is announced. Institutions focus overwhelmingly on format compliance—adopting the new message standard, upgrading the interface, passing the connectivity test. And the harder, less visible question—whether the underlying data is actually ready for the new standard—gets deferred until it becomes a crisis.

I watched this pattern unfold with the original SWIFT messaging standards in the 1990s, when institutions raced to adopt MT formats without addressing the quality of beneficiary information flowing through them. I watched it again with the SEPA migration in the 2000s, where IBAN adoption was treated as the finish line while the address and remittance data remained inconsistent. I observed it with real-time payment schemes, where institutions celebrated technical go-live while discovering that their data could not sustain the straight-through processing rates the new rails demanded.

Each time, the institutions that distinguished between technical compliance and genuine data readiness were the ones that emerged stronger. They processed faster. They screened more accurately. They resolved exceptions in hours rather than days. The difference was never the technology. It was the data discipline.

We are in the same cycle now with ISO 20022 and structured postal addresses.

The migration is real. The deadlines are firm. By November 2026, unstructured postal addresses will no longer be accepted in SEPA payments or SWIFT cross-border messages. Over 3.9 million CBPR+ messages are processed daily on the new rails. The format transition is, in many respects, well advanced.

But as I observe the current state of preparation across the industry, I see a familiar and concerning gap. Institutions are compliant with the format. They are not ready with the data. Translation from MT to MX has largely preserved the formatting syntax, but the underlying data quality has not kept pace. Legacy free-text address content is being carried, essentially unchanged, into new structured message fields.

There is, however, a critical difference from previous cycles. The technology to solve this problem—to transform unstructured addresses into fully structured, compliant, verified output at enterprise scale—now exists and is production-proven. The question is no longer whether it can be done. It is whether institutions will act with sufficient urgency and invest in genuine solutions rather than cosmetic fixes.

This paper examines that question. It draws on published regulatory guidance from the EPC, SWIFT, BIS/CPMI, and PMPG; industry research from EY, Citi, and other institutional sources; operational analysis from production address resolution platforms; and three decades of direct observation across financial services technology.

1. The Regulatory Landscape: Coordinated, Unambiguous, and Imminent

The push toward structured addressing has gained urgency because it is not a single regulatory initiative. It is a coordinated, multi-body mandate with aligned timelines. This coordination is significant. It means there is no jurisdiction to hide in, no standard to defer to, and no ambiguity about direction.

1.1 SEPA: The European Mandate

The European Payments Council has formally mandated that after 22 November 2026, unstructured address formats will no longer be allowed in SEPA payment messages. From that date, only structured or hybrid address data can be used under updated SEPA Credit Transfer and SEPA Instant Credit Transfer rulebooks. This is not guidance. It is a rule change with a fixed effective date.

The EPC has explicitly recommended that institutions use the transition period to move directly to fully structured formats wherever feasible, rather than settling for hybrid as a permanent state.

1.2 SWIFT CBPR+: The Global Network Mandate

SWIFT's Cross-Border Payments and Reporting Plus programme aligns with this timeline. In November 2026, SWIFT will remove unstructured postal addresses from its network as part of Standards Release 2026. From that point forward, all cross-border payment messages must use either fully structured or hybrid addresses, with Town/City Name and Country Code as mandatory fields. Messages failing to meet these requirements will be rejected.

To support the transition, SWIFT has released a free, open-source AI model designed to extract Town and Country from legacy address data. However, the model is limited to these two fields only, does not extract full address components such as street name, building number, or postal code, does not support agent fields, and has acknowledged limitations with non-Latin scripts including Arabic, Cyrillic, and Chinese. It is a helpful bridge tool, but not a comprehensive solution for full address structuring.

1.3 BIS/CPMI: The G20-Aligned Harmonisation

The BIS Committee on Payments and Market Infrastructures, supported by the Payments Market Practice Group, has specified a minimum structured address standard under Requirement 11 of the harmonised data requirements. All ISO 20022 payment messages should include, at minimum, the Country and Town Name in structured fields. Postal Code is strongly recommended. The BIS has framed this as a foundational enabler of efficient, inclusive cross-border payments, positioning it as a priority within the G20 payments roadmap.

1.4 FATF and Wolfsberg: The Compliance Dimension

The global shift to structured address data aligns with broader anti-money laundering and transparency frameworks. The Financial Action Task Force, through Recommendation 16 (Travel Rule), calls for accurate, meaningful originator and beneficiary information in payments, and encourages structured formats. The Wolfsberg Group's Correspondent Banking Due Diligence Questionnaire increasingly expects structured originator data to support effective screening and risk-based monitoring. Structured address adoption is therefore a direct enabler of global regulatory compliance across multiple frameworks.

1.5 The Combined Effect

Together, these mandates represent a change as significant as the earlier adoption of IBAN or BIC standards. By late 2026, financial institutions globally must handle structured address information in payments. There is no opt-out. The outcome is binary: compliant or rejected.

2. The Gap Between Format and Fitness

This is where three decades of observation sharpens into a specific concern. I have seen, repeatedly, how institutions conflate format compliance with data readiness. They are not the same thing.

2.1 Format Migration Does Not Equal Content Migration

Simply switching to the ISO 20022 message format does not guarantee data quality. In practice, unstructured or poor-quality address content can persist inside the new formats if not actively corrected. An XML message might formally accommodate structured tags, but if the underlying customer address remains a single line of free text, the benefit is lost.

Early reports from the transition period confirm this. Even large institutions still grapple with hybrid addresses and truncated data, indicating that legacy free-text inputs are being carried, essentially unchanged, into the new fields. SWIFT's own programme lead at a major European bank has cautioned that addresses captured at the source often remain unstructured, leading to manual intervention, delayed payments, and higher compliance risk even after the format change.

2.2 What "Unstructured" Actually Means Operationally

Unstructured addresses are payment message fields where all address information is captured as a single free-text blob, typically using the ISO 20022 AdrLine field. The same AdrLine might contain street names, city, postal code, or country in varying orders, with no semantic separation.

This makes parsing, interpretation, and validation highly unreliable across the payments chain. Such ambiguity is a primary blocker to straight-through processing, automated compliance screening, and effective data analytics.

2.3 A Taxonomy of Recurring Failures

Industry observation and PMPG guidance reveal a consistent pattern of failure modes:

- **Missing mandatory components:** Town and Country are often incomplete, misplaced, or absent in structured fields, even where required for hybrid formats.
- **Duplication across fields:** Country or Town values frequently appear in both structured elements and AdrLine fields, creating validation conflicts and screening ambiguity.
- **Hybrid format misuse:** Hybrid addresses frequently exceed permitted AdrLine limits, populate AdrLine with data meant for structured fields, or violate character restrictions.
- **Placeholder abuse:** Invalid values such as "N/A" or "NOTPROVIDED" are found in TownName or Country fields, causing payment repairs and screening errors.
- **Embedded financial identifiers lost or corrupted:** Unstructured address fields frequently contain financial identifiers that have been concatenated with postal

address data — IBANs, account numbers, sort codes, BIC/SWIFT codes, payment reference numbers, and other transaction-critical information. These identifiers are routinely misclassified as address components, truncated to fit field-length constraints, or discarded entirely. The consequence is direct: payments fail validation, reconciliation breaks, and straight-through processing rates collapse.

- **Downstream dependency:** Institutions frequently depend on intermediaries or clearing systems to fix address issues, deferring failures and creating delays.

These are not edge cases. They are systematic patterns observable across the industry, resulting in high exception rates, manual investigations, regulatory risk, and poor customer experience.

3. The Cost of Inaction: Quantifying the Problem

The gap between format compliance and data readiness has measurable financial and operational consequences.

3.1 The Scale of Manual Intervention

Industry analysis estimates that up to 60% of cross-border payments still require manual intervention due to address-related issues. Each manual intervention carries direct cost. Studies indicate that 2–5% of all cross-border payments prompt formal investigation, with each case requiring 15–45 minutes of analyst time. At scale, across the global payments network, this represents an annual cost to the industry estimated at $15–25 billion.

3.2 What These Costs Represent

- Operations teams manually repairing payment messages that fail structured field validation
- Compliance analysts investigating false-positive sanctions alerts triggered by ambiguous address data
- Correspondent banks exchanging clarification queries over incomplete beneficiary information
- Customers experiencing delayed payments because their address data could not be processed automatically
- Institutions absorbing reputational cost from processing delays in an era where real-time settlement is the expectation

Every month that institutions defer investment in genuine data quality, these costs compound. They are a tax on operational inadequacy—one that effective address structuring technology can largely eliminate.

3.3 Why the Problem Persists

Despite clear standards and available structured fields in ISO 20022, the persistence of unstructured address data is rooted in systemic constraints:

- **Legacy customer master systems** still store address data in free-text blobs, with no field-level separation for town, country, or postal code.
- **Corporate ERP interfaces** often capture addresses as single-line or semi-structured fields, which pass unchanged into payment instructions.
- **Onboarding and channel gaps** mean that customer onboarding processes often do not enforce address structuring, especially for legacy accounts.
- **Validation gaps in hybrid use** mean that payments with invalid hybrid formatting may pass through the network, deferring failure to downstream parties.
- **End-to-end weak links** in correspondent chains mean that even if the sending bank structures correctly, downstream truncation can reintroduce ambiguity.

This is a systemic problem, not a technical one. And systemic problems require systemic solutions—not point fixes at the message layer.

4. Why Conventional Approaches Fall Short

This is where the practical reality confronts institutional planning. Having established that structured address data is mandatory and that the cost of poor data is substantial, the natural question is: how do we fix this? The answer matters enormously, because not all approaches are equal—and several that appear reasonable on the surface fail under the specific conditions of payment address structuring.

Over the past two years, I have observed institutions attempt four broad categories of remediation. Each has significant limitations that are not always apparent during initial evaluation.

4.1 Regex and Dictionary Lookups: The Brittleness Problem

The most common first instinct is to build rule-based parsers: regular expressions that identify patterns, combined with dictionary lookups that match known city names, country codes, and postal formats. This approach works for clean, well-formatted addresses from a single country. It fails catastrophically when confronted with the real-world complexity of global payment data.

The fundamental limitation is that regex has no mechanism for contextual disambiguation. Consider the word “Paris.” A dictionary lookup identifies it as a city name. But which Paris? Paris, France (population 2.1 million). Paris, Texas (population 25,000). Paris, Ontario. Paris, Tennessee. There are more than eleven cities named Paris across multiple countries. Without understanding the surrounding context—the postal code, the country indicators, the address structure conventions of the originating jurisdiction—a rule-based system cannot determine the correct interpretation.

The same problem multiplies across thousands of similar ambiguities:

1. **“Washington”** could be Washington State, Washington D.C., or Washington Street. A regex pattern cannot distinguish them.
2. **“London”** could be the city in the United Kingdom, London Road (a street name), or a surname. Without context, the parser guesses.
3. **“Cuba Street, Wellington, New Zealand”** contains a sanctioned country name as part of a street address. A dictionary lookup flags “Cuba” as a country. A context-aware system recognises “Cuba” + “Street” as a street name in New Zealand—no sanctions relevance.

The result of regex-based approaches is a high rate of both false rejections (good data flagged incorrectly) and false accepts (bad data passing through unchallenged). In sanctions screening, this distinction has regulatory consequences. Misclassifying “Cuba Street” could mean blocking a legitimate transaction or, worse, missing a genuine sanctions risk because the screening system is overwhelmed with noise from misclassified address components.

Rule-based systems also struggle with the sheer scale of global address variation. There are 195 countries, each with distinct address formatting conventions. Street numbers appear before the street name in the United States, after the street name in Germany, and may not exist at all in Japanese block-based addressing. A single set of regex rules fails in three out of four countries. Building and maintaining rules for 195 country configurations is a multi-year effort that most institutions cannot sustain.

4.2 Generic Large Language Models: The Wrong Tool for the Job

The emergence of large language models has prompted some institutions to consider general-purpose AI as a solution for address parsing. On the surface, this is attractive: LLMs are capable of processing natural language text and can extract structured information from unstructured inputs.

In practice, generic LLMs are fundamentally unsuitable for production payment processing, for reasons that become apparent under operational scrutiny:

- **Non-deterministic outputs:** An LLM may produce different structured outputs for the same input address on successive runs. In payment processing, where auditability and consistency are non-negotiable, this is disqualifying. Regulators and auditors require that the same input always produces the same output with the same rationale.
- **No guaranteed schema compliance:** LLMs generate free-form text responses. They cannot guarantee that their output will conform to a specific ISO 20022 schema with the correct XML tags, field lengths, and mandatory element population. Post-processing to force compliance adds complexity and introduces new failure points.
- **No explainability or audit trail:** When an LLM assigns "London" to the TownName field, it cannot provide a traceable evidence chain explaining why. In a regulatory environment where every sanctions screening decision must be explainable, a "black box" classification is a compliance risk, not a solution.
- **Latency and cost:** General-purpose LLMs typically operate at latencies of 500 milliseconds or more per request, at costs 100 to 500 times higher than purpose-built processing engines. For institutions processing millions of transactions daily, these economics are prohibitive and incompatible with real-time payment SLAs.
- **Data sovereignty risk:** Sending payment data containing personally identifiable information to external cloud-hosted LLM services raises data sovereignty concerns that many institutions and regulators will not accept.

This is not a criticism of LLMs as a technology class. They are powerful tools for many applications. But production payment address processing requires determinism, guaranteed schema output, full explainability, sub-50-millisecond latency, and data sovereignty. Generic LLMs provide none of these.

4.3 Postal Validation Services: Solving the Wrong Problem

A third approach institutions consider is leveraging established postal address validation services—the providers that verify whether an address is a valid postal delivery point. These services are well-established and accurate for their intended purpose.

However, postal validation and payment address structuring are fundamentally different problems:

- **Postal valid does not equal payment valid:** A postally valid address may still fail ISO 20022 structured field requirements. Postal validation confirms the address exists; it does not restructure it into CBPR+ or SEPA-compliant fields with TownName, Country, PostalCode, StreetName, and BuildingNumber in their correct XML elements.
- **No entity extraction or disambiguation:** Postal validators return a binary valid/invalid result. They do not extract entities from the address text, do not distinguish between a street name containing a sanctioned country name and an actual country reference, and do not provide the contextual intelligence needed for sanctions screening.
- **No message-level compliance:** Payment addresses require compliance with specific message schemas (CBPR+, HVPS+, SEPA, CHAPS) that change with each standards release. Postal validation services are not designed to track or enforce these payment-specific requirements.
- **No adaptive learning:** Most postal validation services operate on static rule sets. They do not learn from exception patterns, do not adapt to new address formats, and do not improve over time based on the specific address challenges of the payments domain.

Postal validation is valuable as one component of an address quality strategy. It is not, by itself, a solution to the structured address challenge that the November 2026 mandate requires.

4.4 Manual Remediation and Consultancy Projects: The Timeline Trap

The fourth approach—and perhaps the most dangerous in the current context—is treating address structuring as a manual data remediation project, typically executed through internal teams or external consultancies.

I have observed this approach repeatedly across technology transitions, and the failure pattern is consistent:

- **Unbounded scope:** Address data issues follow an iceberg pattern. The visible problems represent 10–20% of the total. As remediation progresses, teams discover three to five times more dirty data than initial assessments suggested. Scope grows, timelines extend, budgets inflate.
- **Timeline mismatch:** Industry experience suggests manual address remediation projects typically require 13–23 months from initiation to completion. With approximately ten months remaining until the November 2026 deadline, many institutions cannot execute a manual approach within the available window.
- **No learning or leverage:** Manual remediation follows a linear scaling model: twice the data requires twice the work at twice the cost. There is no leverage, no

learning effect, and no efficiency gain over time. Each new batch of addresses starts from scratch.

- **No lasting asset:** Manual projects produce a one-time cleanup. They do not address the ongoing flow of new unstructured data entering the institution through customer onboarding, corporate interfaces, and correspondent channels. Within months of project completion, the remediated data begins degrading as new unstructured content enters the system.
- **Cost escalation:** The typical trajectory is familiar: a project budgeted at a given level reaches "80% complete"—the easy 80%—then discovers the remaining 20% contains the complex edge cases that consume disproportionate time and cost.

> *The executive nightmare scenario: a project budgeted at $2 million over six months that reaches $5 million at fourteen months, declared "80% complete," with the November 2026 deadline approaching. I have seen this pattern in every major data remediation cycle. The deadline does not move.*

4.5 The Common Thread: Why All Four Approaches Are Insufficient

What these four approaches share is a fundamental mismatch between the nature of the problem and the capability of the solution. Payment address structuring is not a simple formatting exercise. It is a complex natural language processing challenge that requires:

- Understanding that "Cuba Street" is a street name, not a country reference
- Knowing that street numbers appear after street names in Germany but before them in the United States
- Recognising that محمد (Arabic), Mohammed, Muhammad, and Mohamed are transliterations of the same name, and that 黃, Wong, Wang, Huang, and Hwang are transliterations of the same Chinese character
- Handling 195 distinct country address formatting conventions, each with regional variations
- Producing deterministic, schema-compliant, explainable output at sub-50-millisecond latency
- Maintaining a complete audit trail for every parsing decision, ready for regulatory inspection

No regex engine, generic LLM, postal validator, or manual project can deliver all of these requirements simultaneously. The problem demands purpose-built technology—and that technology now exists.

5. The Case for Intelligent Address Resolution

If conventional approaches are insufficient, what does a genuine solution look like? This section describes the technology class that has emerged to address the structured address challenge—and why it represents a practical, deployable path forward for institutions facing the November 2026 deadline.

5.1 What "Address Resolution" Means

The term is deliberately chosen. This is not address validation (confirming an address exists). It is not address formatting (rearranging text into a template). It is address resolution: the intelligent transformation of unstructured, ambiguous, multi-lingual address text into fully structured, verified, schema-compliant output—with contextual disambiguation, entity extraction, and a complete evidence trail.

The distinction matters. Validation tells you whether an address is correct. Resolution takes an address that may be incomplete, ambiguous, or malformatted and transforms it into a structured, compliant, verified output—fixing the problem rather than merely flagging it.

5.2 The Technical Requirements

Based on operational observation and the specific demands of payment processing, effective address resolution platforms must meet a precise set of requirements:

- **Deterministic processing:** The same input must always produce the same output. This is non-negotiable for regulatory compliance and auditability. Every parsing decision must be reproducible.
- **Context-aware natural language processing:** The platform must understand that "Cuba Street" is a street name and "Cuba" alone may be a country. This requires natural language processing trained specifically on payment address data, not generic text processing. It must handle sanctioned names appearing in street addresses, city names shared across multiple countries, and entity names embedded in address fields.
- **Guaranteed schema-compliant output:** Output must conform to specific ISO 20022 message schemas—CBPR+, HVPS+, SEPA, CHAPS—with correct field population, character limits, and mandatory element presence. The platform must guarantee compliant output, not produce text that requires post-processing.
- **Multi-script, multi-language capability:** Addresses arrive in Latin, Arabic, Cyrillic, Chinese, Japanese, and dozens of other scripts. The platform must handle transliteration across 50+ standards, recognising that the same person, city, or street may appear in multiple script representations.
- **Sub-50-millisecond latency:** Payment processing operates at millisecond timescales. Address resolution must complete within 50 milliseconds at the 95th percentile to integrate into real-time payment flows without introducing latency.

- **Full explainability and audit trail:** Every parsing decision must be accompanied by an evidence citation explaining why each element was classified as it was, with confidence scores at each stage. This is a regulatory requirement, not a desirable feature.
- **Data sovereignty:** Payment data contains personally identifiable information. The platform must operate on-premise or within the institution's virtual private cloud, with air-gapped deployment as an option. No PII should leave the institution's control.

5.3 The Real-World Challenges: What Makes This Hard

Address resolution at enterprise scale confronts challenges that are not immediately obvious from a regulatory compliance perspective but become critical during implementation. These are the practical difficulties that I have observed separate effective solutions from inadequate ones:

Global Format Variation

There is no universal address format. The position of the building number relative to the street name varies by country. The structure of postal codes varies from four-digit numeric (many countries) to six-character alphanumeric (United Kingdom) to no postal code at all (some jurisdictions). The hierarchy of address elements—whether district appears before or after city, whether prefecture is equivalent to state or province—differs across 195 countries. Japan uses a block-based system with no street names in many areas. Germany places building numbers after street names. The United Kingdom uses named buildings that may or may not include a number. An effective address resolution platform must encode all of these conventions and apply them contextually.

Transliteration Complexity

Cross-border payments routinely involve addresses in non-Latin scripts that must be transliterated for processing. A single Chinese character (黃) can be romanised as Wong, Wang, Huang, or Hwang depending on the dialect and transliteration standard. The Arabic name محمد has over a dozen common English spellings. The German city München appears as Munich, Muenchen, and multiple other variants. An effective platform must recognise all standard transliterations as equivalent and resolve them to a canonical form—across more than 50 transliteration standards.

Financial identifier extraction and preservation: Payment address fields routinely contain embedded financial identifiers — IBANs, account numbers, sort codes, reference numbers, and BIC/SWIFT codes — concatenated with postal address data. The platform must recognise these identifiers within unstructured text, extract them accurately, and route them to the correct output fields without corruption or loss. Misclassifying an IBAN as a street name or truncating a reference number to fit an address field has direct consequences for straight-through processing and reconciliation. This capability requires domain-specific intelligence: understanding which character patterns constitute financial identifiers in which jurisdictional contexts, and preserving them as distinct data elements alongside the structured address output.

Entity Disambiguation in Address Context

Perhaps the most operationally consequential challenge is distinguishing entities embedded within address text. When the word "Iran" appears in an address field, is it the country (sanctions-relevant), a company name ("Iran Trading Co, London, UK"—entity-relevant), or a street name ("Iran Street, Dubai"—location-relevant)? Each interpretation triggers a different operational response. Getting this wrong has direct sanctions compliance consequences.

Effective disambiguation requires deep domain knowledge: understanding that "Cuba" followed by "Street" is a location indicator, that "Havana, FL" refers to a small town in Florida while "Havana" alone likely refers to the Cuban capital, and that "Burma Road, Singapore" is a colonial-era street name, not a reference to Myanmar. This knowledge cannot be approximated by generic text processing. It must be built from decades of exposure to real payment address data across all corridors.

The Maintenance Burden

Address formats, postal codes, country configurations, and regulatory schemas change continuously. New address conventions emerge. Postal code systems are updated. SWIFT releases new schema versions. Sanctions lists evolve. An address resolution capability is not a build-once asset. It requires continuous maintenance of 195 country configurations, transliteration tables, disambiguation patterns, and regulatory rule packs. This ongoing maintenance burden is a critical factor that institutions must assess when evaluating build-versus-buy decisions.

5.4 Production-Proven Technology Exists

The requirements described above are demanding. But they are not theoretical. Purpose-built address resolution platforms meeting these specifications are in production today, processing billions of transactions annually across major global financial institutions. These platforms have accumulated decades of encoded payment address intelligence—disambiguation patterns learned from real investigation cases, transliteration mappings refined across 50+ standards, and country configurations tested against production payment flows from 195 countries.

The operational evidence from these deployments is significant. Institutions using purpose-built address resolution technology report straight-through processing rates exceeding 98%, reductions in address-related false positives of 30–70%, exception handling cost reductions from $25 per case to under $5, and resolution times dropping from 15 minutes of analyst time to under 2 seconds of automated processing.

These are not vendor marketing claims. They are operational metrics from production deployments at institutions that collectively represent a meaningful share of global cross-border payment volume.

5.5 Integration: Where Address Resolution Fits

A critical practical advantage of modern address resolution technology is its flexibility of deployment. Purpose-built platforms are designed to integrate at any point in the payment processing chain:

- **At the point of capture:** Structuring addresses in customer onboarding forms, corporate API submissions, and file uploads before they enter the payment system. This is the highest-leverage intervention: clean data at the point of entry eliminates downstream exceptions.
- **During legacy data remediation:** Profiling, transforming, and structuring historical address records in KYC databases, customer master systems, and transaction archives. AI-powered processing with confidence scoring enables high-volume remediation with human-in-the-loop review for low-confidence outputs.
- **In the payment processing pipeline:** Intercepting and structuring address data in payment messages before they are submitted to SWIFT, clearing systems, or domestic payment rails. This is the critical compliance checkpoint for the November 2026 deadline.
- **As a pre-processor for sanctions screening:** Providing structured, disambiguated address data to sanctions screening engines, reducing the noise that generates false positives and improving the precision of genuine alert detection.
- **In shadow mode for assessment:** Operating alongside existing payment infrastructure without disruption, processing a copy of live traffic to establish baseline metrics and demonstrate accuracy before taking on production responsibility.

The integration model is typically a REST API operating at sub-50-millisecond latency, deployable on-premise or in a virtual private cloud, requiring no changes to core banking or payment systems. This means institutions can deploy address resolution technology rapidly—within weeks rather than months—without the risk and disruption of a core system upgrade.

6. Hybrid Addresses: The Bridge, Not the Destination

The hybrid postal address model was introduced in November 2025 to support the industry's migration from unstructured to fully structured formats. In hybrid form, a payment message includes mandatory TownName and Country in structured fields, alongside up to two lines of AdrLine for remaining detail.

This is pragmatic and sensible. But it carries a risk I have observed in every transitional standard: the temporary becomes permanent.

6.1 The Risks of Normalising Hybrid

- **Inconsistent implementation:** Institutions interpret hybrid differently. Some misuse AdrLine length, fail to populate TownName correctly, or exceed permitted field counts.
- **Silent failures:** Many infrastructures currently defer validation, meaning malformed hybrid addresses pass undetected—but later cause screening failures, delays, or rejections at downstream institutions.
- **Increased compliance risk:** When addresses are partly unstructured, screening engines may misinterpret or overlook critical information, leading to false positives or missed alerts.
- **Accumulating technical debt:** Institutions that normalise hybrid defer cleanup work, embedding structural debt in client records and creating friction in future upgrades and analytics initiatives.

6.2 The Recommended Stance

The policy position this paper advocates is clear: *fully structured by default; hybrid as a fallback; unstructured eliminated.* Industry estimates suggest that every 10% increase in structured address adoption reduces address-related exceptions by 1–2% and screening false positives by 3–5%. With purpose-built address resolution technology available to transform data at any point in the processing chain, there is no operational reason to accept hybrid as a permanent state. The technology to achieve fully structured output exists. The economic case is clear. The question is institutional will.

7. Beyond Compliance: The Strategic Value of Structured Addresses

The compliance mandate is clear. What is less widely understood is the breadth of strategic value that structured address data unlocks. The institutions that will benefit most from this transition are those that see it not as a regulatory burden but as an infrastructure upgrade with compound returns.

7.1 Improved Straight-Through Processing

Clean, structured addresses significantly reduce the need for manual intervention. When all required address fields are present and standardised, payment systems can automatically route and settle transactions without repair. Industry analysis indicates that with fully structured address data, over 98% of cross-border payments can be processed straight-through, largely eliminating manual exception handling. The result is faster payments, lower cost, and better customer experience.

Critically, this includes preserving financial identifiers — IBANs, account references, sort codes — that are frequently embedded in unstructured address fields. When these identifiers are lost or corrupted during structuring, the downstream payment fails regardless of how accurately the postal address itself has been parsed.

7.2 Fewer False Positives in Compliance Screening

Structured data allows screening systems to filter by context, checking city and country fields independently rather than searching unstructured text for keyword matches. Research from EY and Citi indicates that structured party data can reduce false positive alerts by 25–30% through more precise entity matching. For large institutions, this reduction translates directly to lower operational cost and better risk management.

Structured addressing also directly supports compliance with wire transfer regulations — EU Regulation 2015/847, FATF Recommendation 16, and the UK Wire Transfer Regulations — which require complete, accurate originator and beneficiary information in fund transfers. When address data is structured, institutions can programmatically verify that the content obligations of these regulations are met, shifting wire transfer regulation compliance from reactive exception handling to proactive, automated assurance.

7.3 Reduced Investigations and Operational Exceptions

By enforcing structured addresses with mandatory town and country, many exceptions currently caused by incomplete or mis-formatted address information can be eliminated. The structured ISO 20022 model replaces generic exception messages with specific camt.* messages, enabling faster and more automated exception handling. Fewer investigation cases mean lower costs, faster resolution, and better customer experience.

7.4 Better Reconciliation and Treasury Outcomes

Structured address data improves internal reconciliation and reference matching. For corporate treasury teams, this means fewer ambiguities when matching payments to accounts, faster settlement, more accurate working capital forecasting, and reduced Days Sales Outstanding. The richer, standardised data model promotes end-to-end automation and improved reconciliation accuracy.

7.5 Analytics, AI Readiness, and Fraud Detection

Fully structured address information opens new possibilities in analytics, artificial intelligence, and fraud detection. Structured PostCode and TownName fields enable geo-risk scoring. Linking parties with common addresses can detect mule networks or shell companies. Structured data feeds clean inputs to AI and analytics tools, enabling analysis that messy text cannot support. Industry research indicates that 30–40% of AI initiative failures trace to underlying data quality issues. Structured addressing removes one of the largest data quality obstacles in the payments domain.

7.6 Entity Resolution and Cross-Domain Interoperability

Structured addresses support accurate entity resolution by enabling field-level comparison across systems. Golden records—clean, unified views of counterparties—require structured addresses to distinguish between name variants, locations, and hierarchies. The BIS/CPMI data harmonisation agenda positions structured addresses as part of a broader shift to machine-readable party data across KYC, onboarding, ESG, logistics, and more.

8. Implications for Decision-Makers

8.1 For Executive Management

The structured address transition is not a technology project. It is an enterprise data quality initiative with direct implications for operational cost, regulatory risk, customer experience, and competitive positioning.

Executive management should understand three things. First, the compliance deadline is firm with no extension mechanism. Second, the cost of unstructured address data is already being absorbed across the institution but is often invisible because it is distributed across operational teams. Third, production-proven technology exists that can resolve this challenge within the available timeline—the constraint is no longer technical capability but institutional decision-making speed.

8.2 For Chief Information Officers and Chief Technology Officers

Technology leaders face a specific implementation choice. The systems that capture, store, transmit, and validate address data span the entire technology estate. The key technical insight is that address structuring is not a message-layer fix. It requires changes at the point of data capture, data storage, validation logic, and screening configuration.

However, the availability of purpose-built address resolution technology, deployable as a REST API at sub-50-millisecond latency with no core system changes, fundamentally changes the implementation calculus. Shadow mode deployment allows institutions to assess accuracy on live traffic before committing to production. On-premise and VPC options address data sovereignty requirements. The path from evaluation to production can be measured in weeks, not years.

8.3 For Compliance and Financial Crime Leaders

Compliance leaders should view structured addressing as a direct enabler of screening precision and regulatory transparency. Purpose-built address resolution technology with full explainability and evidence trails transforms compliance from a reactive function—investigating false positives after they occur—into a proactive one—eliminating the data ambiguity that generates false positives in the first place. Research indicates reductions in false positives of 25–30% with structured data inputs. For institutions processing millions of transactions, this represents significant savings and, critically, better risk detection.

8.4 For Operations Leaders

Operations leaders bear the most immediate impact of poor address data. The operational case for intelligent address resolution is straightforward: higher straight-through processing rates reduce manual work volume, improve Mean Time to Resolve, and lower per-transaction cost. Industry evidence suggests STP rates above 98% are achievable with fully structured data, compared to rates of 40–60% that many institutions experience today. Operations leaders should advocate for upstream solutions—structured capture at source and automated resolution at every processing stage—rather than accepting the current model of downstream repair.

9. The Implementation Path: A Practical Framework

9.1 Capture at Source

The journey to fully structured addresses begins at origination. APIs, portals, and file-based uploads should enforce ISO 20022-compliant structure, with guided forms, dropdown selections, real-time validation, and auto-suggestion. Country-specific address patterns should be embedded through adaptive forms. Where intelligent address resolution technology is deployed at the capture layer, even free-text entries can be transformed into structured output in real time, before they enter the payment system.

9.2 Remediate Legacy Data at Scale

Most institutions hold years of historical address data in free-text format. The first step is profiling: quantifying data that fails minimum presence requirements. The second step is automated transformation using purpose-built address resolution technology with confidence scoring, segmenting high-quality output from data requiring human review. The third step is governance: establishing ongoing quality controls to prevent re-introduction of unstructured content. The critical point is that AI-powered address resolution enables remediation at scale —millions of records processed with sub-second turnaround, confidence-scored, and audit-trailed—rather than the linear scaling of manual approaches.

9.3 Validate Early, Not at the Network Boundary

As PMPG guidance explicitly warns, infrastructure acceptance does not guarantee correctness. Institutions must embed validation upstream:

- Enforce structured field rules: reject messages where TownName is missing or duplicated in AdrLine.
- Reject non-compliant hybrid patterns: flag messages with more than two AdrLines or improper duplication.
- Deploy address resolution as a pre-send checkpoint: transform and validate address data before submission to SWIFT, clearing systems, or domestic rails. This is analogous to pre-send sanctions screening and should be treated with the same operational rigour.

9.4 Governance and Accountability

Structured address readiness must be governed across functions. Policy must define what qualifies as permissible hybrid use, which channels are exempt only by exception, and what escalation paths exist. Data quality must be assigned explicit ownership across data management, compliance, and operations, with clear metrics and reporting. Structured address strategy must be embedded within enterprise data governance.

9.5 Metrics and Measurement

Institutions should build dashboards that track:

- Structured versus hybrid usage by customer segment, channel, and geography
- STP rate, reject and repair rate, and false-positive screening rate
- Exception handling cost and Mean Time to Resolve
- Hybrid-to-structured transition curves, published internally to drive adoption

Industry benchmarks suggest a cost per exception of $25–50 today, with automated address resolution reducing this by 80–90%. These metrics should be embedded in quarterly operational risk reporting.

10. Looking Forward: 2027 and Beyond

By late 2027, structured addresses will be the default. Validation will be enforced closer to the network edge. Market infrastructures will reject malformed messages. The question for institutions is not whether this future arrives, but whether they arrive prepared.

10.1 Structured Data as Innovation Platform

- **Digital identity and golden records:** Clean address data enables confident entity resolution and Customer 360 use cases.
- **Fraud and sanctions intelligence:** Geo-pattern recognition and false-positive suppression through enriched, structured location data.
- **Real-time exception handling:** Structured input enables automated triage and root-cause analysis.
- **Cross-domain interoperability:** The BIS/CPMI data harmonisation agenda targets machine-readable party data across KYC, onboarding, ESG, and logistics by end-2027.

10.2 Six-to-Twelve Month Outlook

1. **Accelerating enforcement:** Market infrastructures will reject non-compliant messages more aggressively as the deadline approaches.
2. **Visible differentiation:** Institutions with structured data capabilities will show measurable STP and screening advantages.
3. **Technology consolidation:** The market for address resolution will consolidate around platforms offering comprehensive, multi-geography coverage with high confidence scoring and sub-second processing.
4. **Regulatory tightening:** Regulators will use structured data capability as a marker of institutional readiness.
5. **Ecosystem pressure:** Correspondent banks will increasingly require evidence of structured data capability as a condition of service quality.

Conclusion: From Compliance Burden to Transformation Catalyst

Over thirty years in financial services technology, I have learned to look for a specific signal when evaluating whether a technology transition will create lasting value: the width of the gap between stated ambition and actual implementation.

In the structured address transition, that gap is wide. The ambition is clear. The regulatory mandate is firm. The benefits are well-documented. But the reality on the ground, as of February 2026, is that most institutions have achieved format compliance without data readiness.

This is the pattern I have observed in every major technology migration. It is also the pattern that creates the largest opportunity for institutions willing to invest in substance rather than appearance.

There is, however, a critical difference this time. The technology to close the gap exists. Purpose-built address resolution platforms are in production at major global institutions, processing billions of transactions with demonstrated accuracy, at speeds compatible with real-time payment processing, with the explainability and audit trails that regulators require. The question is no longer “can this be done?” It is “will we act in time?”

The institutions that will emerge strongest from the 2026 transition are those that see structured addressing not as a regulatory checkbox but as a foundational data quality investment. They will deploy intelligent address resolution at every stage of the processing chain: at capture, during remediation, in the payment pipeline, and as a pre-processor for screening. They will process faster, screen more accurately, resolve exceptions in seconds rather than minutes, and build analytical capabilities that their peers cannot replicate.

Structured by default. Hybrid as fallback. Unstructured eliminated.

That should be the operating principle for every institution navigating this transition. Not because regulators demand it—though they do. But because the technology to achieve it exists, the economic case is compelling, and the institutions that act will be measurably better, faster, and more resilient than those that do not.

The opportunity is not merely to meet the 2026 deadline. It is to arrive ready for what comes after.

Appendix A: Key Compliance Timelines

Milestone	EPC (SEPA)	SWIFT CBPR+	BIS/CPMI
Coexistence Begins	Mar 2023	Mar 2023	—
Hybrid Grace Period	Nov 2025–Nov 2026	Nov 2025–Nov 2026	—
End of Unstructured	22 Nov 2026	Nov 2026	—
Fully Structured Target	2027+	2027+	End-2027

Appendix B: ISO 20022 Address Elements

Element	XML Tag	Status & Notes
Town Name	TwnNm	Mandatory for structured and hybrid formats
Country	Ctry	ISO 3166-1 alpha-2 code. Mandatory for all formats
Postal Code	PstCd	Strongly recommended by BIS/CPMI
Street Name	StrtNm	Strongly encouraged for fully structured
Building Number	BldgNb	Strongly encouraged for fully structured
Building Name	BldgNm	Strongly encouraged for fully structured
Floor	Flr	Strongly encouraged for fully structured
PO Box	PstBx	Strongly encouraged for fully structured
Room	Room	Strongly encouraged for fully structured
Town Location Name	TwnLctnNm	Strongly encouraged for fully structured
District Name	DstrctNm	Strongly encouraged for fully structured
Country Sub Division	CtrySubDvsn	Strongly encouraged for fully structured
Address Line	AdrLine	Max 2 lines in hybrid; phased out in unstructured

Appendix C: PMPG Hybrid Address Compliance Checklist

Rule	Correct	Incorrect
Town and Country	Populated in structured fields	Empty or placeholder values
AdrLine Usage	≤2 lines, 70 chars each	>2 lines or >70 characters
De-Duplication	No repeats across fields	Town/Country in both AdrLine + structured
Placeholder Values	None present	"N/A", "NOTPROVIDED", etc.
Field Population	Correct elements in correct fields	Street name in TownName field

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About the Author

Parth Desai brings over 30 years of experience at the intersection of artificial intelligence and financial services. He has led the design, development, and deployment of AI-driven systems across major financial institutions, with deep expertise in payments infrastructure, compliance technology, and data quality. This paper reflects his analysis of the structured address transition and its implications for the industry.

Disclosure

The author is CEO of ioNova, which provides address resolution technology for financial services. This paper presents the author's independent analysis of industry trends, regulatory requirements, and technology approaches. While the author's experience informs the analysis, the paper is intended as an objective assessment of the structured address transition and does not constitute product promotion. Institutions should conduct their own evaluations of available technologies.

Disclaimer

This white paper is provided for informational purposes and represents the author's analysis as of February 2026. It does not constitute legal, regulatory, or compliance advice. Data points cited are drawn from publicly available industry sources, institutional research, and operational analysis from production deployments, as noted.

Sources and References

This paper draws on published guidance and research from: European Payments Council (EPC) SEPA Rulebooks; SWIFT CBPR+ Standards and Address Guidelines; BIS Committee on Payments and Market Infrastructures (CPMI) Harmonised Data Requirements; Payments Market Practice Group (PMPG) Address Guidance; EY/SWIFT ISO 20022 Impact Studies; Citi Treasury and Trade Solutions Research; Financial Action Task Force (FATF) Recommendation 16; Wolfsberg Group CBDDQ Standards; McKinsey Global AI Survey; and operational metrics from production address resolution deployments at major global institutions.